GHG Statement Report

Version 1.6

Information requirements for verification are as follows:

- GHG Statement Report (this document) containing information relating to the GHG Statement
- GHG Statement calculations (provided separately)
- Supporting information including copies of raw data used

An appropriate level of information should be provided with a clearly referenced and transparent audit trail of decision making, assumptions, explanations, such that a verifier can trace all inputs, outputs and decision making. All data sources and assumptions must be clearly referenced, transparent and traceable.

After this document has been completed for the first time, it may be acceptable to refer back to relevant sections for subsequent Reporting Periods, for example if there have been no changes to the GHG Assessment scope, data collection methodology, calculation methodology or assumptions.

There is no word limit for answers in sections below.

Section A – General Project Information

General Information	
Name of practitioner who prepared the GHG Statement, and relevant competencies. This should include an overview of relevant qualifications and experience undertaking GHG assessments.	Max Lavine Charm Industrial Protocol and Verification Lead
Name of Project / link to Project page on Isometric Registry	Charm Kansas Bio-Oil Geologic Storage
Date of report	8/19/25
Please specify the Reporting Period this GHG Statement has been prepared for (the Reporting Period describes the time period over which the carbon removal activity assessed in the GHG Statement occurred)	6/1/25-7/31/25
Relevant project information if not included in PDD	

Details and links to any supporting documentation	All raw data and supporting documentation has been submitted either 1) directly to Isometric via API, or 2) directly to the VVB
Please confirm that the GHG Statement has been undertaken in accordance with the information provided in the project boundary section of the PDD. Provide any additional information relating to the GHG Statement project boundary here.	Confirmed

Section B – Project Details

B. GHG Statement - Project Details

Provide information on the Project design and implementation concerning the reporting period.

This should include the following information as a minimum:

- *Key features of the Project including a technical description*
- *information on the implementation status of the project, For the description of the installed technologies, technical processes and equipment, include diagrams, where appropriate*

Vaulted Injection Interruption and Updated Embodied Emissions Estimate

During the course of this Reporting Period, Vaulted paused injections in order to support facility upgrade work. As a result, only one injection was completed in June, and the Vaulted embodied emissions estimate has been updated to include the sitework completed and equipment installed during the upgrade.

Charm Aqueous Bio-Oil Injections and Allocation

Charm was approved by KDHE and Vaulted to begin injecting the “aqueous” fraction of our bio-oil, which is a byproduct of the pyrolysis process. Prior to approval, Charm did not allocate pyrolysis production emissions to the aqueous fraction of the oil to ensure that 100% of emissions were allocated to material that had no injection approval risk. As a result, aqueous bio-oil produced before June of 2025 had no production emissions allocated to the material. From June on, aqueous bio-oil will be included in the allocation of production emissions along with the “oily” bio-oil fraction and bio-char. The methodology of allocating production emissions to pyrolysis products based on each product’s proportion of that Production Period’s carbon mass has not changed.

Standardized Loss Estimate for Bio-Oil Totes

During the course of operations, Charm has measured over 50 totes of bio-oil before and after emptying into sparging batches in order to account for loss. Using this dataset, Charm has computed a standardized tote loss estimate of 5.33kg/tote, which is 125% of the dataset mean. This estimate allows for us to support scaling with greater operational efficiency, while

ensuring bio-oil accounting estimates are conservative.

Section C – Project Changes

C.1 Summary of changes

Please provide details of any changes or deviations from the planned implementation, operation, or monitoring of the project activity that occurred during this reporting period. Project Proponents must determine whether the changes are temporary deviations or permanent changes.
Temporary deviations could include deviations from:

a) The monitoring plan mentioned in the approved PDD.
b) The applied Protocol.
c) The applied modules.
d) Data collection procedure.

If there were any temporary changes or deviations corresponding to this reporting period, please include the details of those changes in this section. If the changes do not fall into the above categories, please note them in Section F of the corresponding PDD instead.

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

Charm-Produced Bio-Oil Injections – Production Period MRV

Charm's integration of Ledger functionality for our CO production facility is not yet complete at the time of this report. Emissions for Charm oil have been computed and reported using the same spreadsheet-based method as outlined in the previous GHG Statement Report.

C.2 Impact of changes

For each deviation, provide a description of the nature, extent and duration of the non-conforming reporting period, and:

a) Propose alternative monitoring arrangements for the non-conforming reporting period; or
b) Apply the most conservative values approach

(If there were no deviations during the reporting period, this section is not applicable. In that case, kindly leave this section blank and mention "Section is intentionally left blank.")

Table: 1 Temporary changes to the project

Period of change	*Nature of change*	*Impact of change*
Current Reporting Period	Use of spreadsheets, rather than Ledger API, to compute carbon intensity of Charm-produced bio-oil	De Minimis

Section D – GHG Statement Methodology

D.1 GHG Statement Methodology - Data

Provide information on the GHG Statement approach and methodology in relation to data collection.

This should include the following information as a minimum:

- *Data collection procedures - how data was requested and gathered, who supplied each data point, the format data was received in, and the date data points were received*
- *How data quality was assessed and how the data hierarchy was followed in terms of using measured, calculated and estimated data. Evidence and justification should be provided in any cases where measured data was not used*
- *Details of data validation processes followed, including treatment of missing data*

Evidence of all raw data that informed the assessment (including data that informed any assumptions) must be appended to the PDD, or linked within the GHG Statement.

Please see corresponding section in the PDD Appendix

D.2 GHG Statement Methodology - Calculations
Provide information on the GHG Statement approach and methodology in relation to calculations. *This should include the following information as a minimum:* ● *General description of the methodology, criteria and procedures used as a basis for the assessment, including reference to any documentation (including Isometric protocols and modules), guidance, industry standards and best practice that were followed* ● *Information on calculation procedures followed* ● *Information on any tools used as part of the assessment* ● *Procedure for selecting emission factors including how age (age of data, and the period over which they have been collected), geography (the region or country from where the data have originated), technology (whether the data are specific to a particular technology or mix of many), methodology (the approach applied to gather or calculate the data) and competency (proficiency of entity that developed the data) were considered.* ● *Assumptions and limitations - provide full transparency in terms of value-choices, rationales and expert judgements* ● *Details of a sensitivity analysis, how parameters were assessed and evidence behind choices* ● *Details of uncertainty analysis* ● *Details of any uncertainty adjustments (e.g. %) applied in instances of high uncertainty*
Please see corresponding section in the PDD Appendix

Section E – GHG Statement Results

GHG Statement Results		
The following information should be provided in this section: ● *Baseline results reported in t CO_2e for the Reporting Period* ● *Net CO_2e removals results reported in t CO_2e for the Reporting Period. These must be aggregated and also broken down into GHG SSRs* ● *If it has not already been covered in the* 'Uncertainty assessment' section of the PDD, provide *outcomes of sensitivity and uncertainty analyses including a statement of how uncertainty affects the results and how it has been addressed to minimize misrepresentation* ● *A statement of how the assumptions and choices made in the assessment are conservative* *The GHG Statement calculations must be provided separately, including all raw data and evidence. The GHG Statement calculations must be clearly referenced with a transparent audit trail of evidence, decision making, assumptions, explanations.*		
	Kg	**MT**
Gross Bio Oil Sequestered	553599	553.599
	Kg CO2e	**MT CO2e**
Gross CO2e Sequestered	804550.6156	804.551
Project Emissions	**Kg CO2e**	**MT CO2e**
Process Emissions	260612.1031	260.612
Site+Amortized Embodied Emissions	176184.8969	176.185
Uncertainty Discounts	2167.175558	2.167
Emissions Reductions	190356.2772	190.356
Total Emissions for Reporting Period (kg)	438964.1756	
Total Emissions for Reporting Period (MT)	438.96	
Net Removals for Reporting Period (kg)	555942.7173	

Net Removals for Reporting Period (MT)	555.94	

Section F – GHG Statement Version Changes

GHG Statement Version Changes
For each statement resubmission, please specify the new version date (which is the same as the resubmission date) and the updated total of CO_2e tonnes removed.

GHG Statement Versions

Version date	*Total CO_2e tonnes removed*

Document History

Version	**Date**	**Description**
v1.6	June 2024	Revision to include Section F - GHG Statement Version changes, in line with the versioning product experience
v1.5	May 2024	Revision to include section on project changes, and other editorial changes
v1.4	November 2023	Public version